

Empirical Proof Record: REFUTED

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Proof ID: f4824485daecda4dbf01b6281d402a92...

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1. Claim

Kimera's representation-distance (relative divergence of the manifold state: coupling / order_parameter / knowledge_mass) between same-multiset return histories GRADES their real $|\Delta \text{ max-drawdown}|$ (Spearman fidelity) better than a standard order-aware retriever, by a margin: $\text{graded_fidelity_advantage} = \text{fidelity}(\text{kimera_state}) - \text{fidelity}(\text{order_aware}) \geq 0.2$. The prime set-distance is scored alongside as a saturation control.

- **Operationalisation:** For 8 real return multisets, generate 24+2 orderings spanning drawdowns; score each representation by Spearman(pairwise distance, $|\Delta \text{ max-drawdown}|$). Headline = mean over windows of $\text{fidelity}(\text{kimera_state}) - \text{fidelity}(\text{order_aware shingle})$; the prime set-distance fidelity (saturation control) and the order-blind mean-pool ~ 0 floor are reported alongside.
- **Threshold:** $\text{graded_fidelity_advantage} \geq 0.2$ spearman_rho
- **H0:** H0: $\text{advantage} < \text{margin}$ — Kimera's manifold-state divergence does not grade drawdown better than generic order-aware retrieval.
- **H1:** H1: $\text{advantage} \geq \text{margin}$ — Kimera's manifold-state divergence carries graded path-magnitude that generic order-aware retrieval does not.

2. Verdict

REFUTED — observed -0.07657866667550224

graded fidelity — kimera_state=-0.024, order-aware bar=0.053, order-blind floor=0.000; prime control=0.061 (saturation check); advantage=-0.077 vs margin 0.2.

3. Evidence

Pillar	Statistic	Value	95% CI	Library
graded-fidelity	graded_fidelity_advantage	-0.076579	—	scipy spearmanr

4. Pre-registration

- Registered at: 2026-05-24T02:21:12.567268+00:00
- Config hash: 5f8c5da7f8defc3355090fb892506f67...
- Data hash: 0753497b317e32f44850f531b693fd36...

5. Signature

9d9081aeecb3b0012676b7539708a5c81f767b22bf7bc60a6ec6520717b7382a